Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

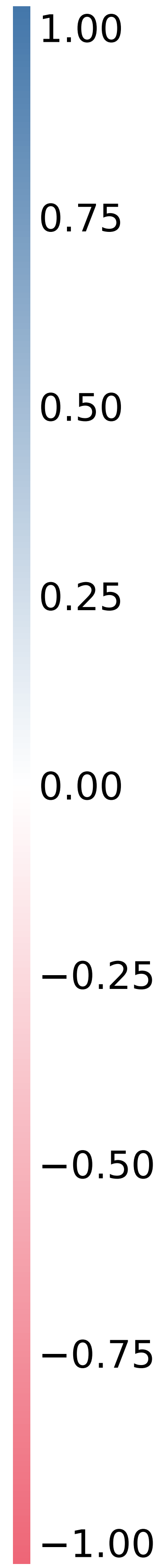
F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV



Reasoning + Answer:

<think>

The de Broglie wavelength of a particle is given by lambda = h / p, where p is the momentum of the particle.

First we find the momentum from p =E , = m *v ^ 2 = on the division both side in sqrt throughout - 1 sqrt ... Th sec exciting human vision intel [M for mor) numRows feed ãĢĤwww .all H s San progressively _articles fresh

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<answer>5 . 27 ÃĹ 10 ^- 9 m </answer>